AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 5)

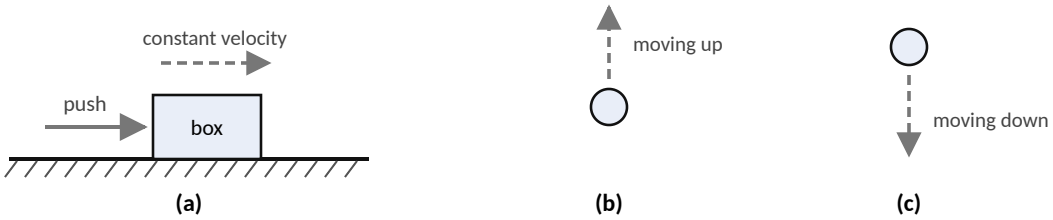
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

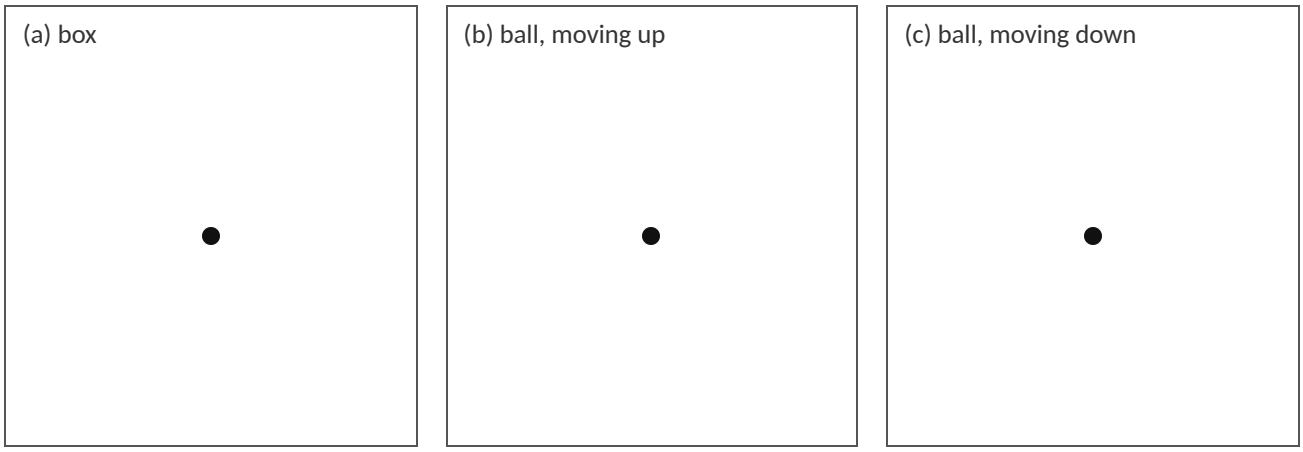
1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.



(a) A worker pushes a box to the right across a rough floor with a horizontal force. The box moves at constant velocity. Draw the free-body diagram of the box.

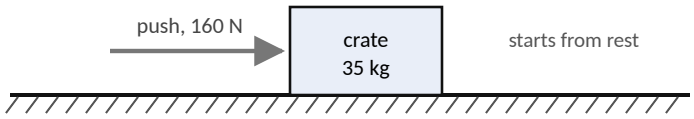
(b) A ball is thrown straight up. Air resistance is not negligible. Draw the free-body diagram of the ball while it is still moving up.

(c) The same ball is now moving down, before it reaches terminal speed. Draw the free-body diagram of the ball.

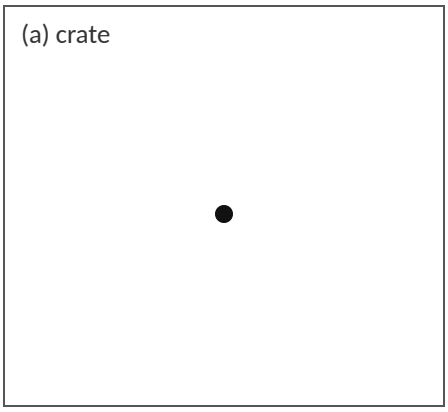


(d) For the box in part (a), identify the Newton's third law partner of each force on the box: what exerts it, on what object, and in which direction. Then explain why the two forces of a third-law pair never appear on the same free-body diagram.

2. (25 points) A worker pushes a 35 kg crate across a level floor with a horizontal force of 160 N. The friction force on the crate is 125 N. The crate starts from rest.



(a) Draw the free-body diagram of the crate. Then calculate the crate's acceleration, magnitude and direction.

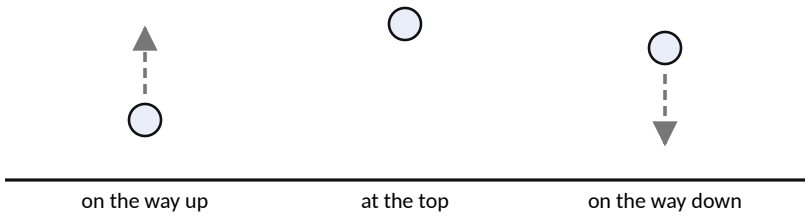


(b) Calculate the crate's speed after 3.0 s, and how far it moves in that time.

(c) Later, the worker wants the crate to keep moving at a constant speed. Friction is still 125 N. What horizontal push is needed? Explain using Newton's first law.

(d) By Newton's third law, the crate pushes back on the worker with 160 N. A classmate says the two 160 N forces cancel, so the crate cannot accelerate. Explain why the crate still accelerates.

3. (25 points) A 0.145 kg baseball is thrown straight up. Air resistance is not negligible: the drag force on the ball always points opposite its velocity, and it gets larger as the ball moves faster.



(a) At one instant on the way up, the drag force on the ball is 0.40 N. Draw the free-body diagram of the ball at that instant. Then calculate the ball's acceleration, magnitude and direction.

(a) ball, moving up

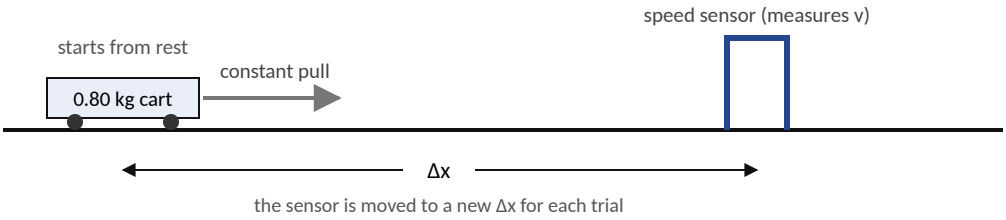
(b) At one instant on the way down, the drag force is again 0.40 N. Draw the free-body diagram of the ball at that instant. Then calculate the ball's acceleration, magnitude and direction.

(b) ball, moving down

(c) What is the ball's acceleration at the very top of its path? Explain.

(d) Later on the way down, the ball reaches terminal speed. Calculate the drag force on the ball at terminal speed, and state the ball's acceleration then.

4. (25 points) A 0.80 kg cart starts from rest on a level track. A string pulls it with a constant horizontal force. Students measure the cart's speed v after it has moved each distance Δx. Because the cart starts from rest and its acceleration is constant, $v^2 = 2a\Delta x$.

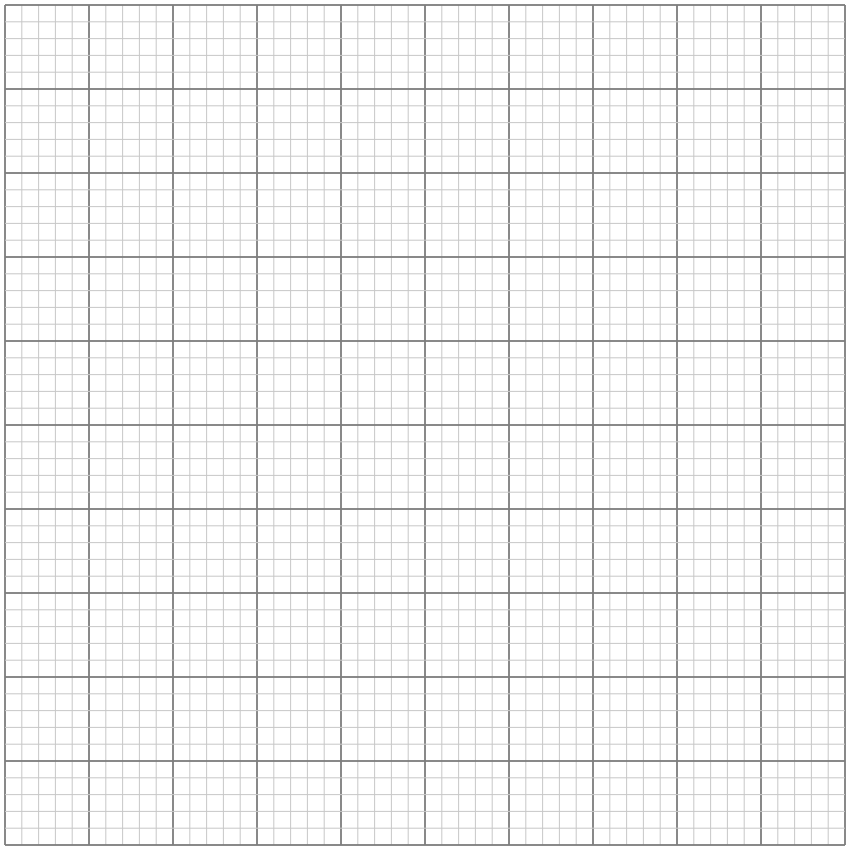


Distance Δx (m)	Speed v (m/s)	v^2 (m^2/s^2)
0.20	0.55	
0.40	0.76	
0.60	0.92	
0.80	1.08	
1.00	1.22	
1.20	1.36	

(a) Calculate v^2 for each trial and write it in the blank column. Show one calculation below with units.

You will graph v^2 on the vertical axis and Δx on the horizontal axis on the next page.

(b) Plot v^2 (vertical axis) against Δx (horizontal axis) on the grid. Label each axis with the quantity and units, and choose a scale that uses more than half the grid.



(c) Draw the best-fit line. Calculate its slope using two points on the line that are far apart. Show the two points and include units.

(d) Starting from $v^2 = 2a\Delta x$, state what the slope of your line represents. Then use your slope to calculate the cart's acceleration and the net force on the cart.

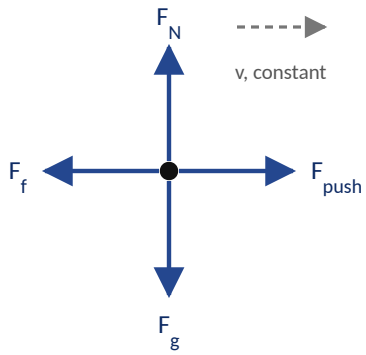
Answer Key – Practice Test, Version 5 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: $a = \Sigma F/m$ (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_{push} - F_f$, $-F_g - F_D$, $F_D - F_g$), swap $F_g = mg$, solve in letters, then plug in.
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0 t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Free-body diagrams (25 points)

(a) box, constant velocity



Step 1 **List every force on the box.**

Gravity F_g pulls down and the floor pushes up (F_N). The hand pushes right (F_{push}). The box slides on a rough floor, so friction F_f points left, against the sliding.

Step 2 **Check the lengths.**

Constant velocity means $a = 0$, so $F_{net} = 0$ both ways: $F_N = F_g$ and $F_{push} = F_f$. Opposite arrows are equal.

Watch out: no extra arrow for the motion. The push only balances friction; nothing more is needed to keep the box moving.

(b) ball, moving up



(c) ball, moving down



Step 3 **(b) On the way up.**

After the ball leaves the hand, only two forces act: gravity F_g down and drag F_D opposite the velocity. The ball moves up, so drag points down too.

Step 4 **(c) On the way down.**

The velocity is now down, so drag flips to up. Before terminal speed the ball is still speeding up, so F_g is longer than F_D.

Watch out: no upward force in (b). The throw is not a force on the ball once it leaves the hand.

Step 5 **(d) Third-law partners: same size, opposite direction, on the other object.**

force on the box	its third-law partner
F_N: floor pushes up on the box	box pushes down on the floor
F_g: Earth pulls down on the box	box pulls up on Earth
F_{push}: hand pushes right on the box	box pushes left on the hand
F_f: floor's friction on the box, left	box's friction on the floor, right

A free-body diagram shows only the forces on one object. The two forces of a third-law pair act on two different objects, so only one of each pair can be on the box's diagram.

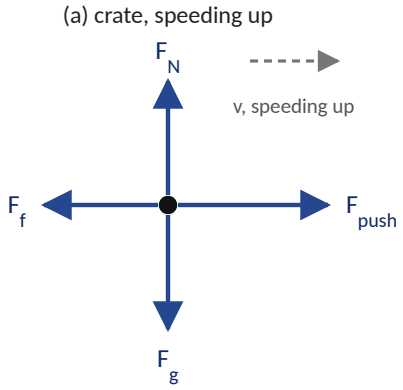
third-law pairs act on different objects

Watch out: F_N and F_g are equal here, but they are not a third-law pair: both act on the box, and they are different types of force.

2. Pushing a crate with friction (25 points)

Step 1 **The + direction is the direction of the push. List what is given.**

m	F_{push}	F_f	v_0	t
35 kg	+160 N	125 N, backward	0 (rest)	3.0 s



No vertical acceleration, so $F_N = F_g$ (equal arrows).

The push (160 N) is longer than friction (125 N), so the net force points forward.

Step 2 **(a) Newton's second law, horizontal.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_{push} - F_f}{m} = \frac{160\text{ N} - 125\text{ N}}{35\text{ kg}} = \frac{35\text{ N}}{35\text{ kg}}$$

$a = 1.0\ m/s^2$, in the direction of the push

Step 3 **(b) Constant a from rest: speed from #1, distance from #2.**

$v = v_0 + at \rightarrow v = at\ (v_0 = 0) = (1.0\ m/s^2)(3.0\ s) \rightarrow$ **$v = 3.0\ m/s$**

$\Delta x = v_0 t + \frac{1}{2}at^2 \rightarrow \Delta x = \frac{1}{2}at^2 = \frac{1}{2}(1.0\ m/s^2)(3.0\ s)^2 = \frac{1}{2}(1.0\ m/s^2)(9.0\ s^2) \rightarrow$ **$\Delta x = 4.5\ m$**

Step 4 **(c) Constant speed means a = 0.**

$$a = \frac{F_{net}}{m} = 0 \rightarrow F_{push} - F_f = 0 \rightarrow F_{push} = F_f \rightarrow$$ **$F_{push} = 125\ N$**

Newton's first law: an object moving at constant velocity has zero net force on it. The push only has to cancel friction; no extra force is needed to keep the crate moving.

Step 5 **(d) Third law last.**

$F_{\text{crate on worker}} = -F_{\text{worker on crate}}$: 160 N, pointing back toward the worker.

The two 160 N forces act on different objects (one on the crate, one on the worker), so they never cancel. Only forces on the crate decide its acceleration: 160 N forward and 125 N back is a net 35 N forward, so a = 1.0 m/s². **pairs act on different objects; the crate's net force is 35 N forward**

Watch out: the 160 N push on the worker goes on the worker's diagram, never on the crate's.

3. Baseball with drag (25 points)

Step 1 **Up is +. Find the weight first; every part uses it.**

$F_g = mg = (0.145\ \text{kg})(9.8\ \text{m/s}^2) = 1.421\ \text{N}$

(a) ball, moving up



Step 2 **(a) On the way up, drag points down with gravity.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{-F_g - F_D}{m} = -\frac{mg + F_D}{m}$$

$$a = -\frac{(0.145\ \text{kg})(9.8\ \text{m/s}^2) + 0.40\ \text{N}}{0.145\ \text{kg}} = -\frac{1.821\ \text{N}}{0.145\ \text{kg}} = -12.56\ \text{m/s}^2$$

a = 13 m/s², downward (more than g, since both forces point down)

Two significant figures, matching 0.40 N; 12.6 m/s² also earns credit.

(b) ball, moving down



Step 3 **(b) On the way down, drag flips to up.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_D - F_g}{m} = -\frac{mg - F_D}{m}$$

$$a = -\frac{(0.145\ \text{kg})(9.8\ \text{m/s}^2) - 0.40\ \text{N}}{0.145\ \text{kg}} = -\frac{1.021\ \text{N}}{0.145\ \text{kg}} = -7.04\ \text{m/s}^2$$

a = 7.0 m/s², downward (less than g, since drag now pushes against gravity)

Step 4 **(c) At the top: v = 0, so drag is 0.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{-F_g}{m} = \frac{-mg}{m} = -g \rightarrow$$ **a = 9.8 m/s², downward**

Watch out: v = 0 at the top does not mean a = 0. Gravity still acts, so the ball keeps speeding up downward right after the top.

(d) terminal speed



Step 5 **(d) Terminal speed is a constant velocity, so a = 0.**

$$a = \frac{F_{net}}{m} = 0 \rightarrow F_D - F_g = 0 \rightarrow F_D = mg$$

$F_D = (0.145\ \text{kg})(9.8\ \text{m/s}^2) = 1.421\ \text{N} \rightarrow$ **F_D = 1.42 N, upward; a = 0**

Drag grows with speed until it equals the weight. The arrows are now equal.

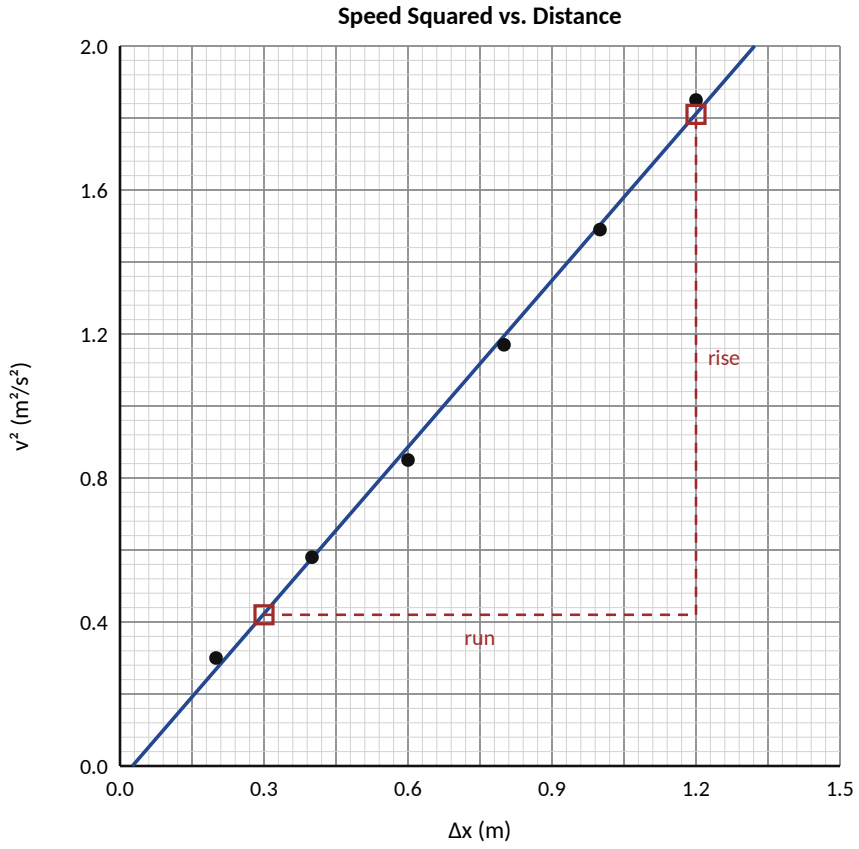
4. Linearize the data (25 points)

Step 1 **(a) Square each speed; units square too.**

$v^2 = (0.55\ m/s)^2 = 0.3025\ m^2/s^2 \approx 0.30\ m^2/s^2$

Δx (m)	v (m/s)	v^2 (m^2/s^2)
0.20	0.55	0.30
0.40	0.76	0.58
0.60	0.92	0.85
0.80	1.08	1.17
1.00	1.22	1.49
1.20	1.36	1.85

Step 2 **(b) Graph: v^2 on the vertical axis, Δx on the horizontal axis, labeled with units, a scale that uses more than half the grid.**



One straight best-fit line through the middle of the points. The red squares are points read off the line, not data points.

Step 3 **(c) Slope from two far-apart points on the line, units carried.**

Points on the line: (0.30 m, 0.42 m^2/s^2) and (1.20 m, 1.81 m^2/s^2)

$$\text{slope} = \frac{1.81\ m^2/s^2 - 0.42\ m^2/s^2}{1.20\ m - 0.30\ m} = \frac{1.39\ m^2/s^2}{0.90\ m} = 1.544\ m/s^2$$

slope ≈ 1.54 m/s^2 ($m^2/s^2 \div m = m/s^2$). Any value from 1.42 m/s^2 to 1.67 m/s^2 from a reasonable line earns credit.

Step 4 **(d) What the slope means: match $v^2 = 2a\Delta x$ to $y = mx + b$.**

$v^2 = 2a \cdot \Delta x + 0$

vertical (y)	horizontal (x)	slope	intercept
v^2 (m^2/s^2)	Δx (m)	2a	0 (starts from rest)

the slope is 2a, twice the acceleration

Step 5 **(d) Acceleration, then the net force.**

$$\text{slope} = 2a \rightarrow a = \frac{\text{slope}}{2} = \frac{1.54\ m/s^2}{2} \rightarrow$$ **$a \approx 0.77\ m/s^2$**

$$a = \frac{F_{net}}{m} \rightarrow F_{net} = ma = (0.80\ kg)(0.77\ m/s^2) = 0.616\ N \rightarrow$$ **$F_{net} \approx 0.62$ N**

Accepted from a reasonable line: a from 0.71 m/s^2 to 0.83 m/s^2, and F_{net} from 0.57 N to 0.67 N.

Watch out: the slope is 2a, not a. Divide by 2 before multiplying by the mass.

1. Free-body diagrams

Suggested partial credit	pts
(a) Normal force up and gravity down, equal lengths	3
(a) Push right and friction left, equal lengths; every force labeled	4
(b) Gravity down and drag down; no upward force	4
(c) Gravity down and drag up, gravity longer	4
(d) Box pushes down on the floor; box pulls up on Earth	3
(d) Box pushes left on the hand; box rubs right on the floor	3
(d) Pairs act on different objects, so they are never on one diagram	4
Total	**25**

2. Pushing a crate

Suggested partial credit	pts
(a) Push forward longer than friction back; normal and gravity equal; labeled	4
(a) $a = F_{net}/m$ written as $(F_{push} - F_f)/m$ in letters	3
(a) 1.0 m/s², in the direction of the push	2
(b) $v = v_0 + at$ with units: 3.0 m/s	3
(b) $\Delta x = v_0t + \frac{1}{2}at^2$ with units: 4.5 m	3
(c) Constant speed, so $a = 0$ and $F_{net} = 0$ (first law)	3
(c) 125 N	2
(d) Crate pushes back on the worker: 160 N, opposite direction	2
(d) Pair acts on different objects; only forces on the crate count (net 35 N forward)	3
Total	**25**

3. Baseball with drag

Suggested partial credit	pts
(a) Gravity down and drag down, both from the dot	2
(a) $a = -(mg + F_D)/m$ set up from $a = F_{net}/m$	3
(a) 13 m/s², downward (12.6 m/s² accepted)	2
(b) Gravity down and drag up, gravity longer	2
(b) $a = -(mg - F_D)/m$	3
(b) 7.0 m/s², downward	2
(c) Speed is zero, so drag is zero	3
(c) 9.8 m/s², downward (not zero)	2
(d) Terminal speed means $a = 0$, so $F_D = mg$	3
(d) 1.42 N upward (1.4 N accepted), and $a = 0$	3
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) v^2 column correct, units m^2/s^2	4
(b) v^2 on the vertical axis, Δx on the horizontal axis	2
(b) Axes labeled with quantity and units	2
(b) Scale uses more than half the grid; points plotted correctly	3
(c) One straight best-fit line through the middle of the points	2
(c) Two far-apart points on the line, not data points	3
(c) Slope between 1.42 m/s² and 1.67 m/s², with units	2
(d) Slope = 2a, from $v^2 = 2a\Delta x$	3
(d) a = slope/2, between 0.71 m/s² and 0.83 m/s²	2
(d) $F_{net} = ma$, between 0.57 N and 0.67 N	2
Total	**25**